

ASSIGNMENT 1

Student Graduation Dataset

Data exploration, cleaning, transformation, splitting and dimensionality reduction

Dataset size	4,424 rows x 35 columns
Predictors / target	34 predictors / Target
Feature groups	17 numerical / 17 categorical predictors
Missing values	0
Duplicate rows	0

Objective

The objective is to prepare the dataset for classification while preserving valid information. The workflow identifies data quality issues, encodes categorical variables, scales numerical features, creates an unbiased train-test split, and evaluates PCA as an optional reduction step.

Key outcome

The data required no missing-value treatment or outlier removal. One-hot encoding expanded the predictor matrix to 245 features; PCA subsequently reduced it to 51 components while retaining 95.22% of the training-set variance.

1. Data Exploration

1a. Structure, summary statistics and data types

Measure	Result
Rows	4,424
Columns	35
Integer columns	29
Floating-point columns	5
Text columns	1
Numerical feature ranges	Age: 17-70; grades: 0-18.875; GDP: -4.06-3.51

Target distribution

Target class	Count	Percent
Graduate	2,209	49.93%
Dropout	1,421	32.12%
Enrolled	794	17.95%

The target has three classes. Graduate is the largest class and Enrolled the smallest, so stratification is used later to preserve this distribution in both data splits.

1b. Missing values and categorical values

No column contains a missing value. The observed categorical values are summarized below.

Categorical feature	Unique	Observed values
Marital status	6	1, 2, 3, 4, 5, 6
Application mode	18	1-18
Course	17	1-17
Daytime/evening attendance	2	0, 1
Previous qualification	17	1-17
Nacionality	21	1-21
Mother's qualification	29	1-29
Father's qualification	34	1-34
Mother's occupation	32	1-32
Father's occupation	46	1-46
Displaced	2	0, 1
Educational special needs	2	0, 1
Debtor	2	0, 1
Tuition fees up to date	2	0, 1
Gender	2	0, 1
Scholarship holder	2	0, 1
International	2	0, 1
Target	3	Dropout, Enrolled, Graduate

2. Data Cleaning

Handling missing values

The dataset contains zero missing values. No imputation, row deletion or column deletion was therefore performed, and all 4,424 observations were retained. Applying imputation when nothing is missing would unnecessarily alter valid data and could introduce bias. Numeric zeros were not interpreted as missing because they have valid meanings, such as zero approved curricular units.

3. Handling Outliers

3a. Detection

Potential outliers were detected in the 17 numerical features with the IQR rule. A value is flagged when it is below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$. IQR was selected because many features are skewed or contain many zeros; unlike a Z-score, it does not assume an approximately normal distribution and is less affected by extreme observations.

Numerical feature	Lower	Upper	Count	%
Application order	-0.500	3.500	541	12.23
Age at enrollment	10.000	34.000	441	9.97
Curricular units 1st sem (credited)	0.000	0.000	577	13.04
Curricular units 1st sem (enrolled)	2.000	10.000	424	9.58
Curricular units 1st sem (evaluations)	0.000	16.000	158	3.57
Curricular units 1st sem (approved)	-1.500	10.500	180	4.07
Curricular units 1st sem (grade)	7.400	17.000	726	16.41
Curricular units 1st sem (without evaluations)	0.000	0.000	294	6.65
Curricular units 2nd sem (credited)	0.000	0.000	530	11.98
Curricular units 2nd sem (enrolled)	2.000	10.000	369	8.34
Curricular units 2nd sem (evaluations)	0.000	16.000	109	2.46
Curricular units 2nd sem (approved)	-4.000	12.000	44	0.99
Curricular units 2nd sem (grade)	6.875	17.208	877	19.82
Curricular units 2nd sem (without evaluations)	0.000	0.000	282	6.37

3b. Decision and justification

The flagged observations were retained. The maximum age is 70, application order is 9, grade is 18.875 on a 0-20 scale, and first-semester evaluations is 45. These values are unusual but possible. Consistency checks found no student with more approved units than evaluations and no student with approved units but a grade of zero.

The large counts for credited units and units without evaluation are mainly caused by zero IQRs: when at least 75% of values are zero, any positive value is flagged. This describes rarity rather than an error. There is therefore no evidence supporting deletion, capping or transformation, which could remove meaningful information about unusual study paths.

4. Data Transformation

4a. Encoding categorical data

Feature type	Treatment	Reason
9 multi-category predictors	One-hot encoding	Codes are labels, not magnitudes or ordered quantities.
8 binary predictors	Keep as 0/1	They already have a valid numerical representation.
Target	Label encoding	Dropout=0, Enrolled=1, Graduate=2 for multiclass classification.

One-hot encoding preserves the nominal meaning of application mode, course, nationality, parental qualifications and occupations. Treating their codes as continuous numbers would incorrectly imply that distances between codes are meaningful. The encoded predictor matrix contains 245 features.

For this fixed educational dataset, pandas `get_dummies` is used for clarity. It observes only feature categories, not target outcomes. In a production system, an encoder would normally be fitted on the training data and configured to handle previously unseen categories.

4b. Feature scaling

The 17 numerical features are standardized to a mean of 0 and a standard deviation of 1. Binary and one-hot indicators remain 0/1. Standardization prevents variables with larger numerical ranges from dominating distance-based or gradient-based models.

Although splitting is discussed in Section 5, it is performed before scaling because `StandardScaler` learns the mean and standard deviation. The scaler is fitted only on training data and then applied unchanged to test data, preventing test information from leaking into training.

5. Data Splitting

5a. Training and testing sets

Set	Rows	Share	Purpose
Training	3,539	80.00%	Fit preprocessing parameters and model
Testing	885	20.00%	Final evaluation on unseen observations

The split uses `random_state=42` for reproducibility and `stratify=y` to preserve the three class proportions.

5b. Importance of splitting

A model learns from the training set, while the test set remains unseen until final evaluation. Testing on new observations provides a realistic estimate of generalization and reveals whether a model has learned training-specific noise. Splitting does not by itself stop overfitting, but it prevents evaluation on the same records used for learning and exposes overly optimistic performance.

6. Principal Component Analysis (Bonus)

PCA was fitted on the scaled training matrix with `n_components=0.95` and then applied unchanged to the test matrix. This retains at least 95% of the training-set variance without using test-set information.

Measure	Before PCA	After PCA
Training shape	3,539 x 245	3,539 x 51
Testing shape	885 x 245	885 x 51
Retained variance	100%	95.22%

Effect on the dataset

PCA reduces 245 encoded features to 51 principal components, a reduction of 79.18%. This can reduce computation, multicollinearity and noise. The trade-off is a 4.78% variance loss and substantially lower interpretability because every component combines many original variables. Prediction quality cannot be inferred from explained variance alone; models with and without PCA would need to be compared.

Conclusion

The dataset is complete and internally consistent enough to retain all observations. Categorical features were encoded according to their meaning, numerical variables were standardized without test leakage, and the stratified 80/20 split produced 3,539 training and 885 testing rows. The final matrices are ready for classification, with PCA available as an optional reduced representation.

Reference

Realinho, V., Vieira Martins, M., Machado, J., and Baptista, L. (2021). Predict Students' Dropout and Academic Success. UCI Machine Learning Repository. DOI: 10.24432/C5MC89.
<https://archive.ics.uci.edu/dataset/697/predict+students+dropout+and+academic+success>